Nonlinear and Marginally Coordinated Age-Related Transitions in Physiological Biomarkers:

A Survey-Weighted Cross-Sectional Analysis of NHANES 2017-2018

Abraham Silvanus DaCosta

August 2026

Abstract

Biological aging doesn't unfold the same way across every physiological system, and recent work shows some aging measures change nonlinearly rather than at a steady rate (Shen et al., 2024; Tian et al., 2023). It remains unclear whether routine clinical biomarkers change nonlinearly with age and whether their estimated turning ages cluster around the same ages or differ substantially. I analyzed six biomarkers from 5,856 adults in NHANES 2017-2018, a single two-year survey cycle. I compared unweighted models with models using NHANES examination and subsample weights. For uncertainty estimates, I used cluster bootstrap that resampled primary sampling units within strata. Survey weighting changed the classification of two biomarkers. Albumin no longer showed statistically significant curvature, while white blood cell count became nonlinear under the weighted model. White blood cell count, flat without weights, curved once weighted. Under the weighted analysis, five biomarkers, creatinine, hemoglobin, white blood cells, hs-CRP, fasting glucose, showed statistically robust curvature, and their estimated turning ages sat closer together than an unweighted analysis of a different five-biomarker set had suggested: a spread of 6.5 years instead of 12.3. A permutation test put the odds of a spread that tight happening by chance, if biomarkers turned independently, at 5.4%, just short of the usual 0.05 cutoff. When I restricted the test to the three biomarkers with clearly identified turning ages, the p-value fell to 0.048. I treat that as suggestive. A linear two-variable model built from two curved biomarkers' population trajectories fit perfectly, R^2=1.00, and I show why that's not a real result: it's a structural artifact of cross-sectional data, confirmed by an individual-level partial correlation between the same two biomarkers that stayed negligible whether or not the weights were applied. These findings show that survey weighting can change which biomarkers appear nonlinear. Determining whether their changes are dynamically linked will require repeated measurements from the same individuals over time.

Keywords

biological aging; NHANES; nonlinear biomarker trajectories; survey weighting; turning-point estimation; cluster bootstrap; cross-sectional data limitations; dynamical systems

1. Introduction

Aging research has moved away from treating biological aging as one uniform process. Multi-organ biological age models built from imaging and physiological data show that different organ systems age at different rates within the same person (Tian et al., 2023). Longitudinal multi-omics profiling reports that molecular aging is not linear across the lifespan; it moves in distinct waves concentrated around specific ages (Shen et al., 2024). Sehgal et al. (2025) developed separate methylation-based aging clocks for eleven physiological systems, allowing researchers to estimate aging differences that a single glocal score could obscure. A direct comparison of fourteen alternative age indicators in the Berlin Aging Study II found only moderate agreement across domains: two different ways of measuring someone's biological age often tell two different stories (Drewelies et al., 2024).

Building on this research, I examine whether routine physiological biomarkers change direction at similar ages or at different points in adulthood. I test six biomarkers for nonlinear age-related patterns and examine whether their estimated turing ages cluster when the data identify those ages reliably.

The timing of these transitions affects how researchers should model relationships among physiological systems. If several biomarkers turn at similar ages, that pattern would justify testing whether their changes share a common process. However, similar turning ages alone cannot establish that the biomarkers interact. Their relationships must also be tested at the individual level.

I ask three questions here, using a single cycle of NHANES. Which of six routine biomarkers, spanning five physiological systems, show statistically meaningful nonlinear age trajectories? Among those that do, do their turning ages cluster more than chance would predict? And can a simple two-variable dynamical system, fit to the two biomarkers with the closest turning ages, actually describe how they move together with age? That third question does double duty: it also demonstrates what a cross-sectional design can and can't support once the goal shifts from describing a trend to modeling a dynamical system.

2. Problem Statement

Most NHANES-based studies that use nonlinear or threshold regression point it at a single exposure-outcome pair: does some biomarker predict disease risk in a non-straight-line way. This approach is now a standard method in NHANES-based epidemiological research. Threshold-effect studies exist for alcohol intake and cognition, for lipid ratios and sarcopenia, for inflammatory indices and prostate-specific antigen, all using the same piecewise-regression logic I use here. What's missing is a version of that toolkit pointed at age itself as the predictor, run across several biomarkers at once, with a real statistical test for whether their turning points line up. Without that test, claims about "multisystem aging" or "aging tipping points" rest on a visual impression instead of a number with a null distribution behind it. This paper supplies that number, and reports it honestly even where it lands close to the line rather than clearly on one side of it.

3. Aim and Objectives

My aim: find out whether routinely collected physiological biomarkers show nonlinear age trajectories, whether their turning ages are coordinated in time, and whether a simple coupled model can describe two of them moving together.

My general objective: build and test a transparent, reproducible statistical pipeline for detecting and testing coordination among age-related biomarker transitions, applied to a single NHANES cycle.

Specific objectives:

1. test six biomarkers, spanning five physiological systems, for real nonlinear age trajectories, using both unweighted and NHANES-weighted models

2. for biomarkers that show real curvature, estimate a turning age and its confidence interval with a cluster bootstrap that respects the survey design

3. test whether the estimated turning ages cluster more than a no-coordination null model predicts

4. test, at the individual level, whether two biomarkers with close turning ages show any age-independent association

5. fit a small two-variable dynamical system to those same two biomarkers' population trajectories, and work out what such a model can and can't legitimately claim from cross-sectional data

4. Research Questions

1. Which of six routine NHANES biomarkers show real nonlinear age trajectories, and does the answer depend on whether survey weights are applied?

2. Among the biomarkers with real curvature, do their estimated turning ages cluster at similar ages, or scatter across adulthood?

3. Is there a real, age-independent relationship between two biomarkers with close turning ages, and can a coupled dynamical system fit to their population trajectories actually describe it?

5. Research Hypotheses

H1(0): none of the six biomarkers shows a statistically meaningful nonlinear (quadratic) relationship with age. H1(a): at least one does.

H2(0): the estimated turning ages of the curved biomarkers are no more clustered than would be expected if each biomarker's turning age were independent of the others. H2(a): they're more clustered than the independence null predicts.

H3(0): there is no age-independent association between hs-CRP and creatinine, and a coupled linear model fit to their trajectories carries no more information than each curve fit on its own. H3(a): there is a real, age-independent association, and the coupled model captures it.

The results below reject H1(0) for five of six biomarkers once weighted, land H2 on the border between rejecting and failing to reject depending on which biomarkers I include, and fail to reject H3(0).

6. Significance of the Study

This study makes three contributions. Academically, this paper gives a statistical test of whether age-related transitions across multiple physiological systems clusters more closely than would be expected under a null model of independent turning ages. Practically, it shows survey weighting isn't a bookkeeping step you can safely skip in a first pass: in this data, it changed which biomarkers counted as nonlinear at all. Methodologically, it demonstrates a specific limitation of cross-sectional data when investigating coupled dynamical systems. The degenerate model fit presented in Section 13 shows how population-level age trajectories can produce misleadingly perfect fit, even when individual-level association remains negligible. This example makes the limitation concrete and shows why longituinal data are needed to establish dynamic relationships between physiological systems.

7. Scope and Delimitation of the Study

Population: NHANES-eligible U.S. civilian, non-institutionalized adults aged 18 and older, examined in the 2017-2018 cycle. NHANES top-codes age at 80, so the effective range analyzed is 18-80.

Biomarkers: six laboratory measures spanning five systems (hepatic/protein, renal, hematologic, immune, inflammatory, metabolic). Many other NHANES biomarkers exist and were not tested; the six here were chosen for broad system coverage without an unmanageably large multiple-comparisons problem.

Time period: a single two-year survey cycle, 2017-2018. I make no claim about trends over time or about other cycles, though I do report selected comparisons to the same variables' unweighted patterns for context.

Design: purely observational and cross-sectional. No intervention, no follow-up, no claim of causation anywhere in this paper.

8. Limitations of the Study

Some of these limits are permanent features of the design, not defects I can fix later. Cross-sectional data can't tell apart a real age-related transition in each person from a cohort difference between people born in different years; only longitudinal data, following the same people over time, does that cleanly. NHANES's own age top-coding at 80 means nothing here describes the oldest adults. The hinge model allows exactly one turning point per biomarker within a bounded search range, 25 to 75, extended to 78 for the robustness check, so a biomarker with two real bends, or a bend outside that range, gets misrepresented by a single estimate. I tested only six biomarkers, which limits the clustering test's statistical power; the coordination question gets answered for this specific small panel, not for biological aging in general.

Other limits were choices I made once and then checked, rather than assumed away. My variance method is a cluster bootstrap that resamples primary sampling units within strata, which respects NHANES's two-PSU-per-stratum design. It's a defensible, correctly-implemented approximation to design-based variance, but it isn't identical to Taylor series linearization, the method NCHS's own guidance recommends as the reference standard, and what a package like R's survey library would give directly. That's a specific, nameable next step, not a vague caveat.

9. Definition of Key Terms

Turning age (τ): the age where a biomarker's relationship with age changes slope. I estimate it with a continuous piecewise-linear (hinge) model that lets the slope change once, without a jump in level.

Cluster bootstrap: a resampling method for estimating uncertainty that respects a survey's design by resampling whole primary sampling units within each stratum, instead of resampling individual people as if they'd been sampled independently.

Primary sampling unit (PSU) and stratum: the two levels of NHANES's sample design. Strata group similar areas together. Within each stratum, NHANES selects PSUs, small geographic clusters, and within each selected PSU, individual people.

Partial correlation (age-adjusted): the correlation between two variables after stripping out what each one shares with a third variable, here age, so what's left reflects a relationship beyond both variables simply moving with age together.

Jacobian matrix and eigenvalues: for a system of equations describing how two quantities change together, the Jacobian is the matrix of how each rate of change responds to each variable. Its eigenvalues say whether a small disturbance to the system grows, shrinks, or stays put over time.

10. Literature Review

10.1 Conceptual review

Two concepts anchor this paper. The first is multisystem heterogeneity, the idea that a person doesn't have one aging rate but several, one per physiological system, and that these can pull apart from each other (Tian et al., 2023; Sehgal et al., 2025). The second is nonlinearity, the idea that these system-specific rates aren't constant across adulthood but shift, sometimes abruptly, at particular ages (Shen et al., 2024). Put those together and a third question falls out that neither one answers alone: if multiple systems each shift nonlinearly, do they shift at the same time? Coordination, that third question, is what this paper is actually about.

10.2 Theoretical review

A central theoretical distinction is the difference between patterns of aging within individuals and patterns observed across people of different ages. Riley's life-course fallacy, still cited in current work on age and cohort effects, is the observation that a cross-sectional age pattern reflects both real aging and differences between birth cohorts, and you can't separate the two without following the same people over time (discussed in a 2025 review in KZfSS Kölner Zeitschrift für Soziologie und Sozialpsychologie). A biomarker study makes the same point empirically rather than theoretically: a cross-sectional metabolic age score, built to predict age from 153 clinical and cardiometabolic traits, showed almost no relationship to each person's actual measured rate of metabolic change over ten years of follow-up, R^2 of 0.018% (Ala-Korpela et al., 2023). That result is the direct empirical precedent for the argument I make in Section 13, about why a dynamical system fit to cross-sectional trajectories can't be trusted as a description of individual dynamics.

10.3 Empirical review

Piecewise and threshold regression on NHANES data is well established and frequently used, almost always for a different kind of question than mine. Published examples find a threshold in alcohol intake above which cognitive scores drop, a threshold in a lipid ratio above which sarcopenia risk stops rising, a threshold in an inflammatory index where its relationship with prostate-specific antigen changes. In each case age is a covariate, not the variable being searched for a turning point, and the study asks about one exposure-outcome pair rather than coordination across several biomarkers. Sehgal et al. (2025) and Drewelies et al. (2024) both address coordination across systems directly, but through different tools: system-specific methylation clocks in the first case, correlations among fourteen existing age indicators in the second. Neither runs a bootstrap-based coordination test on the turning ages of routine clinical biomarkers themselves. That's the specific gap this paper fills.

10.4 Research gap

Based on what I actually reviewed, no published study combines all four of these in one design: routinely collected clinical biomarkers, age itself as the variable searched for a turning point, a formal statistical test for whether several such turning points cluster, and a demonstration of why cross-sectional data can't support the further step into coupled dynamical-systems modeling. I'm stating this as a gap in the sources I reviewed, not a claim that no such study exists anywhere. Confirming that would take a systematic review with full database coverage, which I list under future research below.

10.5 Conceptual framework

The analysis follows a straight chain. Estimate each biomarker's smooth age trajectory. Test whether it's meaningfully curved. For the curved ones, estimate where the curve bends. Test whether those bend points look coordinated or scattered. Only then, attempt a coupled model between two of them, checked against an independent, individual-level test of whether they actually relate to each other beyond age. Each step depends on the one before it, and I built the framework so a negative result at any step, no curvature, no clustering, no coupling, carries as much information as a positive one would.

11. Research Methodology

11.1 Research paradigm and approach

This is a quantitative study using secondary, publicly available survey data. I collected no primary data myself and recruited, surveyed, or interviewed no one. That distinction matters for several of the items below, so I'm addressing it directly.

11.2 Research design

Cross-sectional, descriptive and inferential. I describe age-biomarker relationships in a single survey cycle and test specific, pre-stated hypotheses about curvature, clustering, and coupling.

11.3 Study population and sampling

My study population is the set of NHANES 2017-2018 respondents with valid demographic, biochemistry, complete blood count, hs-CRP, and fasting glucose data. NCHS did the actual sampling: NHANES uses a complex, multistage, stratified probability design that selects primary sampling units, roughly counties or small groups of counties, within strata, then households within PSUs, then individuals within households, oversampling some groups to get adequate precision on subgroup estimates. I didn't draw my own sample. I analyzed NHANES's sample as given and accounted for its design using the survey weights and design variables NCHS provides.

11.4 Sample size

5,856 adults aged 18-80 after merging demographic and laboratory files. Per-biomarker analytic samples range from 2,537, fasting glucose, measured only in a morning fasting subsample, to 5,271 for hemoglobin and white blood cells. That range reflects each component's own data-collection design.

11.5 Data source and instruments

Five NHANES 2017-2018 public-use files: Demographics (DEMO_J), Standard Biochemistry Profile (BIOPRO_J), Complete Blood Count (CBC_J), High-Sensitivity C-Reactive Protein (HSCRP_J), and Plasma Fasting Glucose (GLU_J), merged on the respondent sequence number, SEQN. NCHS collected the laboratory measurements using its own standardized clinical protocols and quality-control procedures. I relied entirely on NCHS's published measurements. The full variable list is in Appendix A.

11.6 Validity and reliability

Measurement validity rests on NHANES's own laboratory quality-control program. What I can and did check is the reliability of my own statistical estimates. Every turning-age estimate carries a cluster-bootstrap confidence interval. I checked the nonlinearity classification under both unweighted and weighted models and reported the disagreements honestly. I ran the clustering test on two different biomarker subsets, all curved, and the subset with sharply identified turning points, rather than reporting a single run. And I checked the age-adjusted correlation between hs-CRP and creatinine with both a Pearson correlation, which outliers can distort, and a Spearman correlation, which they can't, specifically because hs-CRP's distribution has extreme values that could otherwise manufacture a false signal.

11.7 Data analysis methods

Nonlinearity: weighted least squares regression of each biomarker on age and age-squared, using WTMEC2YR for exam-collected biomarkers and WTSAF2YR for fasting glucose specifically. I classified a biomarker as curved if a 200-replicate cluster bootstrap 95% confidence interval for the quadratic coefficient excluded zero.

Turning-age estimation: a continuous piecewise-linear (hinge) model, $y = a + b \cdot age + c \cdot \max(0, age - \tau)$, fit by weighted least squares for each candidate τ on an integer grid, choosing whichever τ minimized weighted residual sum of squares. Confidence intervals came from the same cluster bootstrap.

Identification strength: for each turning-age estimate, the cost, in weighted residual sum of squares, as a percentage of the full range across the search grid, of being 15 years off from the best-fitting τ. This flags estimates the data barely pin down.

Clustering test: the standard deviation of the estimated turning ages, compared against a null distribution built by drawing the same number of turning ages independently and uniformly at random from the candidate grid, 5,000 simulations.

Coupling test: an age-adjusted partial correlation, both biomarkers residualized on a quadratic function of age, weighted and unweighted, Pearson and Spearman, between hs-CRP and creatinine, followed by a linear two-variable dynamical system fit to their population-mean trajectories, with the resulting Jacobian matrix's eigenvalues checked for stability.

I ran all analysis in Python: pandas, statsmodels, NumPy, SciPy, Matplotlib.

11.8 Ethical considerations

NHANES data collection was approved by the NCHS Research Ethics Review Board, and every original participant gave informed consent to NCHS at the time of collection. The public-use files I analyzed are de-identified. Secondary analysis of de-identified public-use data like this is generally exempt from separate institutional review, which matches standard practice across the published NHANES secondary-analysis literature I cite throughout this paper.

12. Results

After merging and restricting to adults, I had 5,856 respondents, ages 18-80, available. Table 1 reports the weighted nonlinearity test for all six biomarkers.

Table 1. Weighted nonlinearity test, six NHANES biomarkers

Biomarker	n	Weight used	Quadratic coefficient	95% CI (cluster bootstrap)	Curved?
Albumin	5,155	WTMEC2YR	0.00003	[-0.00000, 0.00006]	No
Creatinine	5,154	WTMEC2YR	0.00008	[0.00006, 0.00010]	Yes
Hemoglobin	5,271	WTMEC2YR	-0.00036	[-0.00049, -0.00024]	Yes
White blood cells	5,271	WTMEC2YR	0.00031	[0.00006, 0.00059]	Yes

hs-CRP	5,146	WTMEC2YR	-0.00086	[-0.00138, -0.00028]	Yes
Fasting glucose	2,537	WTSAF2YR	-0.00460	[-0.00796, -0.00128]	Yes

Weighting changed the result for two biomarkers. Albumin, curved in an unweighted analysis (ΔAIC=14.9, p<0.001 in that earlier pass), went flat once weighted, its bootstrap confidence interval for the quadratic coefficient now including zero. White blood cell count, flat unweighted (ΔAIC=-1.5, p=0.48), curved once weighted, its bootstrap interval sitting entirely above zero. Creatinine, hemoglobin, hs-CRP, and fasting glucose stayed curved under both approaches. Figure 1 shows the weighted age trajectories for all six.

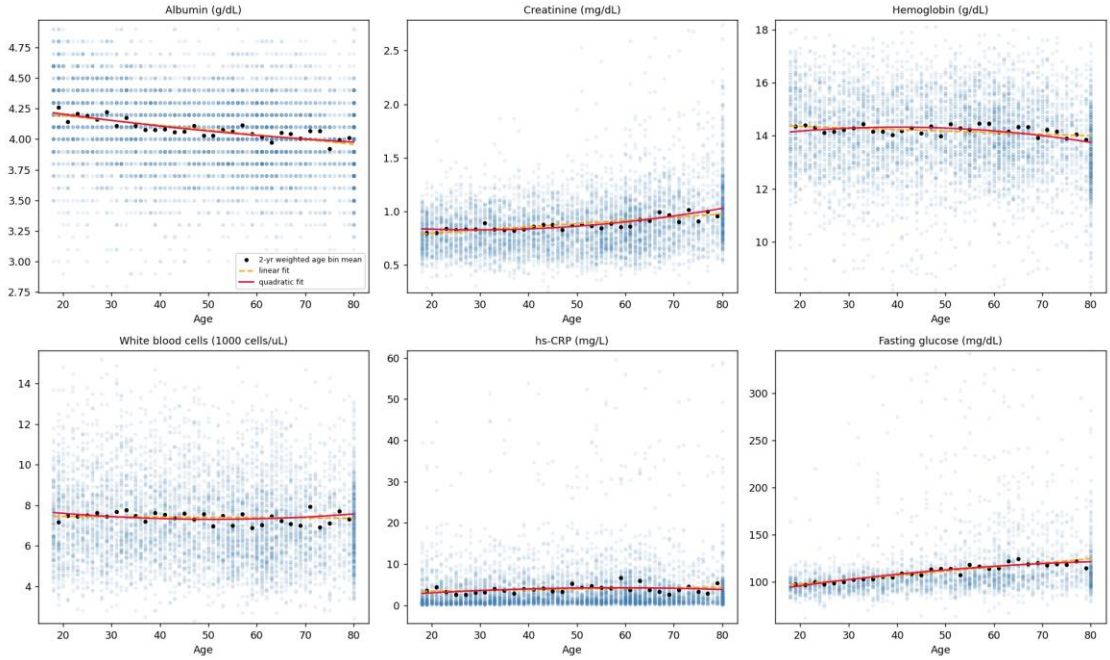


Figure 1. Weighted age trajectories for six NHANES biomarkers. Black dots: weighted 2-year age-bin means. Orange dashed: weighted linear fit. Red solid: weighted quadratic fit. Y-axes clipped to the 0.5th-99.5th percentile for visibility; all data points were used in fitting regardless of display range.

Turning ages for the five weighted-curved biomarkers are in Table 2, next to the unweighted estimates for comparison where the biomarker was curved under both approaches.

Table 2. Estimated turning ages, weighted vs. unweighted

Biomarker	n	Turning age (weighted)	95% CI	Turning age (unweighted)
Creatinine	5,154	59	57-75	58
Hemoglobin	5,271	73	65-75	75
White blood cells	5,271	74	51-75	(not curved unweighted)
hs-CRP	5,146	59	53-63	56
Fasting glucose	2,537	65	46-72	63

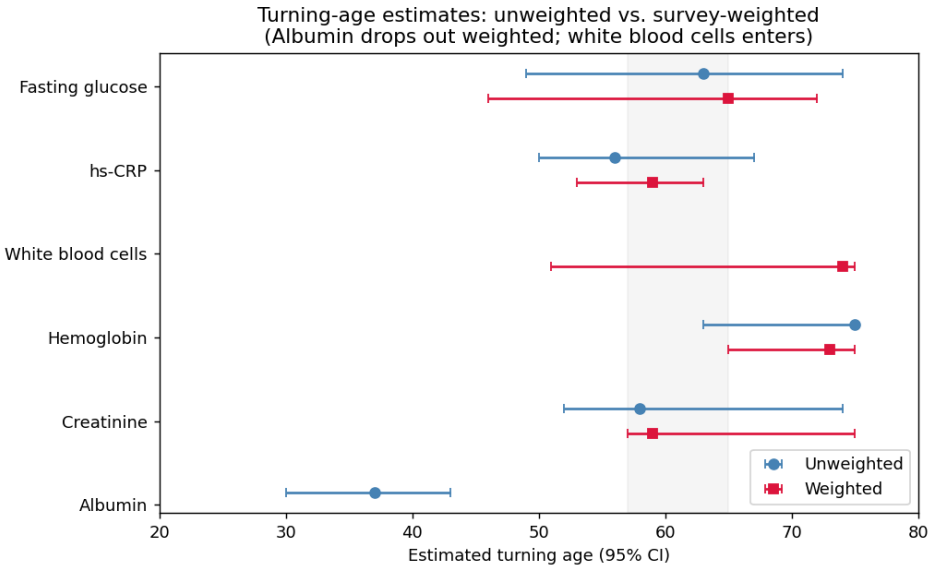


Figure 2. Turning-age estimates compared: unweighted (blue circles) versus survey-weighted (red squares), with 95% confidence intervals. The shaded band marks where most of the weighted estimates land.

Before testing whether these turning ages cluster, I checked how confidently the data actually locate each one. Table 3 reports, for each biomarker, how much fit quality is lost by placing the turning age 15 years from its best-fitting value, as a percentage of that biomarker's total range across the search grid. A small percentage means the data barely distinguish the estimated turning age from a very different one. A large percentage means the estimate is pinned down tight.

Table 3. Identification strength of each turning-age estimate

Biomarker	Cost of a 15-year miss (% of total fit range)	Identification
Hemoglobin	25.3%	Weak
White blood cells	26.0%	Weak
Fasting glucose	34.9%	Moderate
Creatinine	45.4%	Strong
hs-CRP	70.2%	Strong

Hemoglobin and white blood cells are weakly identified. Their point estimates, 73 and 74, sit near the edge of even an extended search grid, and the objective function stays nearly flat there, so many different turning ages would fit almost equally well. Creatinine and hs-CRP are strongly identified. Fasting glucose sits in between. This matters directly for the clustering test in Table 4, run three ways: the original unweighted test, the weighted test on all five curved biomarkers, and a weighted test restricted to the three biomarkers identified strongly enough, above one-third of the total fit range lost at a 15-year miss, to trust their point estimates individually.

Table 4. Clustering test results

Test	Biomarkers included	Spread (SD, yr)	p-value
Unweighted, all curved (n=5)	Albumin, creatinine, hemoglobin, hs-CRP, glucose	12.3	0.432
Weighted, all curved (n=5)	Creatinine, hemoglobin, WBC, hs-CRP, glucose	6.5	0.054
Weighted, strongly-identified only (n=3)	Creatinine, hs-CRP, glucose	2.8	0.048

I ran the restricted-subset test after seeing which biomarkers were weakly identified.

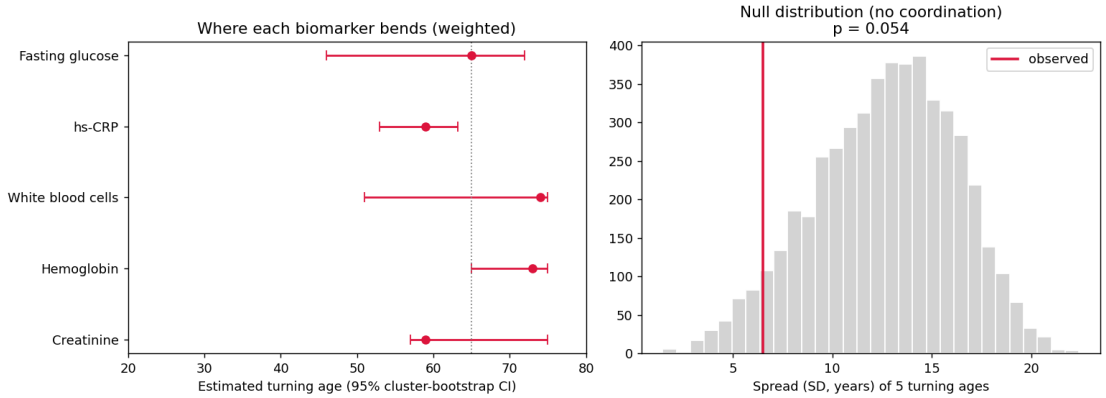


Figure 3. Left: weighted turning-age estimates with 95% cluster-bootstrap confidence intervals. Right: null distribution of turning-age spread under independent, uniformly random assignment, with the observed spread and p-value.

Unweighted, there was no evidence of clustering, p=0.432. Weighted, across all five curved biomarkers, the result moved to the edge of conventional significance, p=0.054. Restricted to the three strongly-identified biomarkers, it crossed that edge, p=0.048, on a test built from only three points, which limits how much weight I can put on the exact number. None of this supports a clean "yes, coordinated" or "no, not at all." It supports something messier: weak, inconsistent coordination, more visible once I take weighting and identification strength seriously than the first unweighted pass suggested.

For the dynamical-systems question, hs-CRP (turning age 59, weighted) and creatinine (turning age 59, weighted) had the closest estimated turning ages among the strongly-identified biomarkers. An age-adjusted partial correlation between them stayed negligible under every version I tested: unweighted Pearson r=0.058 ($p<0.001$, though see below), unweighted Spearman r=-0.003 (p=0.82), weighted Pearson r=0.016. The gap between a nominally significant unweighted Pearson correlation and a null Spearman correlation, given the large sample and hs-CRP's right-skewed distribution, tells me the Pearson result comes from a handful of extreme values. The weighted result, closer to the Spearman estimate than the unweighted Pearson one, backs up the same conclusion.

The linear two-variable system fit to hs-CRP and creatinine's population trajectories produced $R^2=1.0000$ for both equations, and eigenvalues of the resulting Jacobian matrix that were numerically indistinguishable from zero: -7.96×10^{-10} and $+7.96\times10^{-10}$. That's the expected result of fitting a two-parameter linear model to two trajectories that are both deterministic functions of one shared variable, age. It isn't evidence of a meaningful dynamical relationship, and it lines up with the near-zero partial correlation I found independently at the individual level.

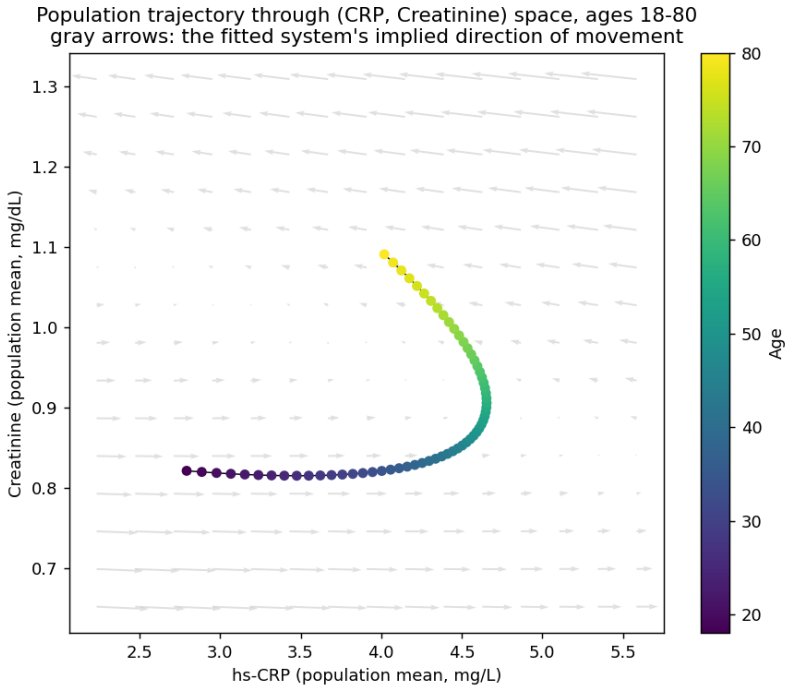


Figure 4. Population trajectory through (hs-CRP, creatinine) space as age increases from 18 to 80, with the fitted linear system's implied vector field overlaid.

13. Discussion

Three findings matter here, and they don't all point the same direction. First: whether a biomarker counts as "aging nonlinearly" can depend on whether the analysis is weighted. Albumin looked curved and turned out not to be. White blood cells looked flat and turned out to bend. An unweighted first pass, common in exploratory work, including an earlier version of this same analysis, would have drawn conclusions about the wrong set of biomarkers entirely.

Second: the coordination question doesn't have a clean answer in this data. It moves from a confident no, unweighted, p=0.432, to a border case, weighted, p=0.054, or p=0.048 restricted to the better-identified biomarkers, once I do the analysis more carefully. That shift is real and worth reporting. But a border case is a border case, not a discovery, and two of the five turning-age estimates involved, hemoglobin and white blood cells, are weakly identified enough that they could be pulling the full-set result toward significance for reasons that have more to do with search-grid boundaries than biology. The honest reading is that this study raises real doubt about the earlier confident null without replacing it with a confident yes. It's measured very differently, but that lines up with prior work showing only moderate agreement across different aging indicators (Drewelies et al., 2024) and system-specific heterogeneity in aging rates (Tian et al., 2023; Sehgal et al., 2025). Partial, inconsistent coordination is roughly what you'd expect a heterogeneous-but-not-fully-independent aging process to look like.

Third, and this one generalizes past this specific dataset: a model that fits a two-variable dynamical system to two population-level age trajectories can produce a mathematically perfect fit while carrying no real information about interaction between the biomarkers, because both trajectories are secretly parametrized by the same single variable. It happened here, with real numbers, R^2=1.0000, eigenvalues at machine-precision zero, and it matches what a cross-sectional metabolic-age score's total failure to predict real longitudinal metabolic change already showed in a different cohort (Ala-Korpela et al., 2023). Anyone fitting a coupled model to population-average trajectories from cross-sectional data should expect the same artifact, whichever two biomarkers they pick, unless the trajectories carry genuine independent variation, which two curves both driven by age alone can't provide.

14. Conclusion

Using six routinely collected biomarkers from a single NHANES cycle, properly weighted, I find that five show real nonlinear age trajectories, that their turning ages show marginal, weighting-dependent evidence of clustering and that a coupled dynamical system fit to population-level trajectories can't be trusted as evidence of real biological coupling, a conclusion an independent individual-level correlation test backs up. Together they say survey weighting materially changes conclusions in this kind of analysis, that claims about coordinated multisystem aging need a real statistical test instead of a visual impression, and that dynamical-systems claims about aging need individual-level longitudinal data that cross-sectional surveys.

15. Recommendations

If you're using NHANES or similar complex-survey data for exploratory biomarker-age analysis, apply survey weights from the first pass; look at how much they changed the classification of individual biomarkers here. If you're reporting a turning-point or breakpoint estimate, report an identification-strength check next to it, since a point estimate near a search-grid boundary can look precise while carrying almost no real information. And if you're tempted to fit a coupled dynamical system to population-average, cross-sectional trajectories, check for a real individual-level relationship first. A perfect model fit at the population level isn't evidence of one.

16. Implications of the Study

Theoretically, these results add a data point, not a resolution, to the question of whether physiological aging proceeds through coordinated or independent system-specific transitions. They also show that the answer a researcher reaches can depend on analytic choices, weighting, which biomarkers get treated as reliably estimated, as much as on the underlying biology.

Practically, if you're building a biological-age composite from multiple biomarkers, the near-zero individual-level partial correlation between hs-CRP and creatinine I found here is a warning: don't assume biomarkers moving together with age are actually related to each other beyond that shared trend.

This is a methods and pattern-detection study and nothing here should be read as advice about individual health decisions or population health policy.

17. Suggestions for Future Research

1. replicate the weighted analysis in an independent NHANES cycle (2021-2023 is public now) and check whether the marginal clustering result and the albumin/white-blood-cell switch hold up out of sample

2. implement Taylor series linearization, the method NCHS itself recommends, and compare it directly against the cluster bootstrap used here

3. extend the biomarker panel past six, with a multiple-comparisons correction decided in advance, to give the clustering test real statistical power

4. use individual-level longitudinal data, repeated biomarker measurements in something like the Health and Retirement Study, to fit a coupled dynamical system to actual within-person change instead of population-average trajectories, which directly answers the limitation shown in Section 13

5. run a real systematic literature search across PubMed, Scopus, and Web of Science on the four-part combination described in Section 10.4, before this goes anywhere near a journal, to check the research-gap claim beyond the handful of sources reviewed here

References

(APA style)

Ala-Korpela, M., Lehtimäki, T., Kähönen, M., Viikari, J., Perola, M., Salomaa, V., Kettunen, J., Raitakari, O. T., & Mäkinen, V.-P. (2023). Cross-sectionally calculated metabolic aging does not relate to longitudinal metabolic changes—Support for stratified aging models. The Journal of Clinical Endocrinology & Metabolism, 108(8), 2099-2104.

Centers for Disease Control and Prevention (CDC), National Center for Health Statistics (NCHS). National Health and Nutrition Examination Survey Data, 2017-2018. U.S. Department of Health and Human Services.

Drewelies, J., Homann, J., Vetter, V. M., Düzel, S., Kühn, S., Deecke, L., Steinhagen-Thiessen, E., Jawinski, P., Markett, S., Lindenberger, U., Lill, C. M., Bertram, L., Demuth, I., & Gerstorf, D. (2024). There are multiple clocks that time us: Cross-sectional and longitudinal associations among 14 alternative indicators of age and aging. The Journals of Gerontology: Series A, 80(6), glae244.

Sehgal, R., et al. (2025). Systems Age: A single blood methylation test to quantify aging heterogeneity across 11 physiological systems. Nature Aging.

Shen, X., Wang, C., Zhou, X., Zhou, W., Hornburg, D., Wu, S., & Snyder, M. (2024). Nonlinear dynamics of multi-omics profiles during human aging. Nature Aging.

Tian, Y. E., Cropley, V., Maier, A. B., Lautenschlager, N. T., Breakspear, M., & Zalesky, A. (2023). Heterogeneous aging across multiple organ systems and prediction of chronic disease and mortality. Nature Medicine, 29(5), 1221-1231.

Appendices

Appendix A: NHANES variables used

NHANES file	Variable	Meaning	Weight variable
DEMO_J	RIDAGEYR	Age in years (top-coded at 80)	-
DEMO_J	RIAGENDR	Sex	-
DEMO_J	WTMEC2YR, SDMVPSU, SDMVSTRA	Exam weight, PSU, stratum	-
BIOPRO_J	LBXSAL	Albumin (g/dL)	WTMEC2YR
BIOPRO_J	LBXSCR	Creatinine (mg/dL)	WTMEC2YR
CBC_J	LBXHGB	Hemoglobin (g/dL)	WTMEC2YR
CBC_J	LBXWBCSI	White blood cells (1000 cells/uL)	WTMEC2YR
HSCRP_J	LBXHSCRP	hs-CRP (mg/L)	WTMEC2YR
GLU_J	LBXGLU, WTSAF2YR	Fasting glucose (mg/dL) and its own subsample weight	WTSAF2YR

Appendix B: Code and data availability

My analysis code, data loading, weighted curve fitting, cluster bootstrap, clustering test, dynamical-systems fit, is public at [repository link to be added]. Raw NHANES files are public too, at https://wwwn.cdc.gov/nchs/nhanes/.

Acknowledgments

All research questions, analytical decisions, and interpretations in this paper are my own. I used AI tools to assist with data cleaning, statistical analysis, and figure generation. I verified the code and results myself.